Minor traumatic intracranial haemorrhage

BIG-1 • Risk stratification, observation and disposition

WHY THIS MATTERS

A small traumatic intracranial haemorrhage on CT is not synonymous with a high risk of deterioration. Historically, many patients were transferred or admitted solely because blood was present. The Brain Injury Guidelines (BIG) instead combine imaging, examination, drugs and clinical context to identify very-low-risk patients.

RECOGNIZE THE LOW-RISK PHENOTYPE

Parameter	BIG-1 example
Neurology	GCS 15; normal examination
Risk modifiers	No intoxication or anticoagulation
Skull	No skull fracture
SDH / EDH	Small collection, typically ≤4 mm
Intraparenchymal	Small, single focus ≤4 mm
SAH / IVH	Trace SAH; no intraventricular blood

Important: These are illustrative BIG-1 criteria. Antiplatelet exclusions, exact haemorrhage definitions and observation requirements differ across implementations. Use the complete locally adopted protocol, not this table alone.

BEDSIDE APPROACH

- Stabilize first: anticoagulants, evolving neurological deficit, seizures, significant mechanism and other injuries override a low-risk disposition shortcut.
- Review the images and report, not just the phrase “tiny bleed.” Document a serial neurological examination.
- Arrange observation, consultant discussion and return precautions according to a validated institutional pathway.

CLINICAL PEARL

The question is not “Is there blood?” It is “What is this patient’s risk of clinically important deterioration?”

WHAT THE NEW PAPER ADDS

Zelt and colleagues systematically evaluated BIG-based management across studies. Among patients categorized BIG-1, pooled adverse events were uncommon; different outcomes had different denominators:

Outcome	BIG-1 rate
Neurological deterioration	4/1,330 = 0.30%
Neurosurgical intervention	1/2,142 = 0.05%
Death	3/1,397 = 0.21%

For identifying patients at risk of neurosurgical intervention, pooled sensitivity was 98.7% (95% CI 97.4–99.4%). Rare events, heterogeneous pathways and variable follow-up limit certainty. This is risk stratification evidence, not a randomized discharge trial.

WHEN NOT TO SHORTCUT DISPOSITION

- Anticoagulant exposure, coagulopathy, abnormal or worsening examination, unreliable assessment or concerning concomitant injuries.
- No safe home observer, poor access to return care, repeated vomiting, severe symptoms, or discordance between imaging and clinical course.
- No locally agreed BIG implementation: discuss with the trauma/neurosurgical service rather than assuming automatic discharge.

ONE-MINUTE REHEARSAL

For a stable patient with trace traumatic SAH and GCS 15, articulate: “Does this patient satisfy every low-risk criterion? What observation is required? Who will observe at home, and can they return promptly?”

EVIDENCE AND READING

Zelt N et al. Brain injury guidelines for the management of traumatic brain injury: a systematic review and meta-analysis. CJEM 2026;28:968–981. DOI: 10.1007/s43678-026-01218-y. Evidence level: systematic review; figures above from the published abstract/newsletter.

VExUS and the congested kidney

Venous hypertension • RV physiology • Venous Doppler

UNDERSTAND THE PHYSIOLOGY

Renal perfusion depends on more than arterial pressure. Raised right atrial pressure can transmit through the venous system, impair renal outflow, increase interstitial pressure and reduce effective filtration. A patient may have adequate MAP and still have poor renal perfusion because venous back-pressure is high.

MENTAL MODEL

Effective renal driving pressure ≈ MAP − renal venous pressure. A patient can be hypotensive AND fluid overloaded.

THE VEXUS EXAMINATION

Combine an enlarged IVC with Doppler evidence of organ venous congestion. A full assessment requires anatomy, angle correction/appropriate sampling and awareness of rhythm and valvular disease.

Vessel	Progression with congestion
Hepatic veins	Normal S>D → reduced S → systolic reversal
Portal vein	Continuous → increasingly pulsatile
Intrarenal veins	Continuous → biphasic → diastolic-only

Severe Doppler abnormalities in multiple vascular beds strengthen evidence of congestion. IVC diameter alone is neither a reliable volume-status test nor a fluid-responsiveness test.

WHERE ULTRASOUND MISLEADS

- Severe tricuspid regurgitation affects hepatic waveforms; atrial fibrillation modifies venous pulsatility.
- Cirrhosis/portal hypertension can alter portal flow; intra-abdominal pressure and ventilation affect the IVC.

PRACTICAL DECISION SEQUENCE

- First scan LV/RV, pericardium and lungs. Identify major RV dysfunction, LVOT obstruction, tamponade or obvious volume loss.
- Ask whether the problem is low forward flow, high backward pressure, or both; examine kidneys, lactate, urine output, lung B-lines and fluid balance.
- If venously congested, avoid reflexive boluses. Treat the cause, support perfusion pressure, and consider decongestion only when haemodynamics permit.
- Repeat the examination after intervention: VExUS tracks physiology, but has not been proven as a stand-alone outcome-improving treatment target.

WHAT THE NEW PAPER ADDS

Peteinidou and colleagues prospectively studied 418 patients admitted with acute heart failure. Persistent venous congestion was associated with subsequent death or heart-failure admission (adjusted HR 1.77; 95% CI 1.17–2.67). The relationship is prognostic and observational: it does not show that VExUS-guided diuresis improves survival.

ONE-MINUTE REHEARSAL

Shock + oliguria + dilated RV + plethoric IVC + pulsatile portal flow. Explain why “AKI equals give fluids” is unsafe; determine the cause of right-sided pressure and whether forward flow needs support.

RESUS BINDER CANDIDATE

The congested kidney: VExUS, RV failure and why low arterial pressure does not equal volume depletion.

EVIDENCE AND READING

Peteinidou E et al. Determinants and Prognostic Value of Persistent Congestion Assessed by Venous Excess Ultrasound Score in Acute Heart Failure. Diagnostics 2026;16:3374. DOI: 10.3390/diagnostics16203374. Prospective observational study.

Sedation during mechanical ventilation

Propofol • Dexmedetomidine • Midazolam • Ketamine • Isoflurane

START WITH A TARGET, NOT A DRUG

The clinical objective is an appropriate level of comfort, synchrony and safety while minimizing hypotension, delirium, immobility and delayed awakening. Use a specified sedation target (commonly light sedation where feasible); provide analgesia and reassess pain, agitation and ventilator interaction.

COMPARE KEY CHOICES

Agent	Practical strengths / trade-offs
Propofol	Rapid titration; hypotension, hypertriglyceridaemia and infusion syndrome risk.
Dexmedetomidine	Light/cooperative sedation; bradycardia and hypotension.
Midazolam	Anticonvulsant; accumulation and slower awakening.
Ketamine	Analgesic, bronchodilating; variable haemodynamics in depleted shock.
Isoflurane	Inhaled titration; specialized device/scavenging and vasodilation.

THE BEDSIDE ALGORITHM

- Correct pain, hypoxaemia, ventilator dyssynchrony, delirium triggers and equipment problems before simply escalating hypnotics.
- Define whether light or deep sedation is truly required. Paralysis requires reliable analgesia and hypnosis; it does not provide either.
- Choose an agent by cardiovascular profile, neurologic requirements, organ dysfunction, anticipated duration and local staffing.

WHERE INHALED AGENTS FIT

Inhaled sedation offers potentially rapid control with relatively little accumulation, but physiological and outcome advantages must be demonstrated. It introduces device, scavenging and environmental considerations; IV rescue drugs may still be required.

SAVE-ICU RANDOMIZED TRIAL

Jerath et al. randomized mechanically ventilated adults with acute hypoxaemic respiratory failure in Canadian ICUs; 780 were included in the primary analysis. Comparing isoflurane-based sedation with usual intravenous sedation:

Hospital mortality	Rate
Isoflurane strategy	42.5%
Intravenous strategy	43.8%
Risk ratio	0.98 (95% CI 0.84–1.14)

There was no significant mortality advantage; ventilator-free and ICU-free days were similar. This does not prove equivalence across all clinical settings and does not argue that IV sedation is inherently superior for every patient.

THREE QUESTIONS BEFORE INCREASING A SEDATIVE

- Is this pain, delirium, dyssynchrony or a true need for deeper sedation?
- Will a higher dose reduce systemic pressure or delay liberation from ventilation?
- Can a reversible cause be fixed instead?

CLINICAL PEARL

The right depth of sedation matters more than the newest sedative.

EVIDENCE AND READING

Jerath A et al. SAVE-ICU randomized clinical trial. JAMA 2026. DOI: 10.1001/jama.2026.18471. Randomized open-label trial (Canadian Critical Care Trials Group).

ARDS: seeing beyond a white chest film

Syndrome recognition • RALE score • Lung-protective priorities

ARDS IS A CLINICAL SYNDROME

Bilateral opacities alone do not diagnose ARDS. Integrate timing, an appropriate clinical insult, oxygenation impairment and imaging findings not explained solely by hydrostatic oedema. ARDS and cardiogenic oedema may coexist.

A SYSTEMATIC IMAGING APPROACH

- Assess the distribution of opacities, symmetry, focality and evolution over time.
- Integrate lung ultrasound: diffuse versus focal B-lines, pleural irregularity, consolidations and effusions.
- Use focused echo to evaluate LV/RV function and alternative or additional hydrostatic causes.

RALE: QUANTIFY RATHER THAN GUESS

The Radiographic Assessment of Lung Edema (RALE) score divides the frontal chest radiograph into four quadrants. Score each quadrant for extent of consolidation (0–4) and density (1–3, when opacities are present). Multiply extent × density and sum the quadrants: total 0–48.

Quadrant	Extent × density
Right upper	0–12
Left upper	0–12
Right lower	0–12
Left lower	0–12

RALE standardizes radiographic burden. It is not a specific test for the cause of pulmonary oedema, nor a substitute for interpreting physiology and clinical context.

WHAT THE STUDY ADDS

Zimatore and colleagues evaluated RALE in a post hoc analysis of 236 mechanically ventilated adults; 92 were adjudicated as ARDS. A RALE threshold of 20 produced:

Diagnostic property	Estimate
Sensitivity	85%
Specificity	83%
AUROC	0.93 (95% CI 0.90–0.96)

The dataset was selected and largely reflects an ICU research context. A threshold derived from these data should not automatically determine ED diagnosis or management.

WHAT CHANGES BEDSIDE CARE?

- Identify and treat the triggering process. Use lung-protective ventilation with appropriate tidal volumes based on predicted body weight and plateau/drive pressures.
- Assess PEEP, oxygenation, haemodynamics, recruitability and RV strain rather than reacting only to an imaging score.
- Escalate to proning and advanced support according to severity, contraindications and established protocols.

CLINICAL PEARL

A score can describe how abnormal the chest X-ray looks; it cannot tell you why the patient has respiratory failure.

EVIDENCE AND READING

Zimatore C et al. Diagnostic Performance of the RALE Score for ARDS. Critical Care Explorations 2026;8:e1495. DOI: 10.1097/CCE.0000000000001495. Post hoc observational analysis.

AI-assisted cardiac POCUS

Agreement is not completeness • Understand the algorithm’s blind spots

THREE DISTINCT TASKS

Image acquisition, measurement and clinical interpretation are not interchangeable. A machine may accurately estimate chamber size yet fail to address the diagnosis that matters most in shock. Before trusting any automated output, know what the model was validated to detect.

A FOCUSED UNSTABLE-PATIENT SCAN

Question	Clinical consequence
Pericardial effusion / tamponade?	Urgent obstructive shock management
RV enlarged or failing?	PE, RV infarct, pulmonary hypertension
LV function / filling?	Low-output versus underfilling
Dynamic LVOTO?	Inotropes can worsen physiology
Valve pathology?	Acute MR/AR can drive shock

CRITICAL DISTINCTION

Agreement: when AI reports a finding, how often does it match expert interpretation?
Yield: how often can it provide an interpretation at all?
Scope: does the system even evaluate the important abnormalities?

DANGEROUS SHORTCUT

“AI says normal” is not equivalent to “no tamponade, RV failure, dynamic LVOTO or major valve disease.”

WHAT THE PAPER STUDIED

Manivel et al. assessed 195 ED echocardiograms across four Australian departments. Expert reviewers evaluated images and compared them with a defined AI interpretation system. For validated parameters:

Metric	Result
Definitive AI interpretation (strict criteria)	46%
Definitive AI interpretation (permissive criteria)	70%
Agreement when both definite	Approximately 76%

The interpretation scope was incomplete relative to expert focused echocardiography. Agreement statistics must be interpreted alongside reader variability, image selection, study population and the distribution of normal/abnormal cases.

HOW TO USE AI SAFELY

- Use automated findings as an aid within a complete clinical examination, never as a rule-out test outside validated scope.
- Save representative clips, inspect image quality and review raw anatomy when the automated report disagrees with physiology.
- Escalate indeterminate or discordant scans to experienced operators or formal echocardiography.

TEACHING EXERCISE

Given a hypotensive patient whose AI report states “normal LV function,” ask: was the RV actually assessed? Was there an effusion? Were Doppler/valve findings reviewed?

EVIDENCE AND READING

Manivel V et al. Artificial Intelligence Echocardiographic Interpretation in the ED. Emergency Medicine Australasia 2026;38:e70364. DOI: 10.1111/1742-6723.70364. Multicentre agreement study.

Venous blood gases in shock

Know which numbers answer the clinical question

WHAT A VBG IS GOOD FOR

A peripheral venous blood gas provides useful information about acid-base state and often trends in CO_2 when paired with clinical assessment. It is not a direct measurement of arterial oxygenation, and peripheral venous oxygen tension is highly dependent on local flow and extraction.

INTERPRET THE COMPONENTS SEPARATELY

Measurement	Practical interpretation
pH / bicarbonate	Useful assessment of metabolic acid-base disturbance
Venous PCO_2	Useful trend; venous–arterial gap varies, especially in shock
Peripheral PvO_2	NOT a substitute for arterial PaO_2 or oxygen delivery
Lactate	Marker to interpret serially with perfusion and context

WHY VENOUS PO_2 IS UNRELIABLE

The blood sampled from a peripheral vein reflects oxygen consumption and blood flow in that local tissue bed. Cold limbs, venous stasis, tourniquet time, reduced regional perfusion and extraction all influence the result. A low PvO_2 does not independently prove systemic hypoxaemia.

CLINICAL PEARL

Do not make an arterial oxygenation or resuscitation decision from an isolated peripheral venous PO_2.

BEDSIDE INTERPRETATION SEQUENCE

- Determine acid-base status: acidemia versus alkalemia; primary metabolic or respiratory process; estimate compensation when appropriate.
- Integrate lactate trend with perfusion examination, mental status, capillary refill, urine output and haemodynamic findings.
- Use pulse oximetry and arterial sampling when accurate arterial oxygenation, $PaCO_2$, or an A–a calculation is required.
- If shock persists, reassess physiology rather than “chasing” a venous oxygen value.

WHAT THE STUDY ADDS

Corbo et al. assessed 973 ED patients who had VBG sampling. Peripheral venous PO_2 did not add useful predictive discrimination for 48-hour deterioration beyond age and lactate.

Prediction model	AUROC
Age + lactate	0.649
Age + lactate + PvO_2	0.654

This is a retrospective prognostic analysis; it does not mean peripheral venous oxygen physiology is unimportant in all contexts. Central or mixed venous saturation is a different measurement with different limitations.

ONE-MINUTE REHEARSAL

If a sick patient has venous PO_2 of 3.5 kPa but normal oxygen saturation, ask what the venous sample represents before ordering oxygen or escalating resuscitation.

EVIDENCE AND READING

Corbo J et al. Association of Initial Peripheral Venous pO_2 with 48-Hour Clinical Deterioration. Academic Emergency Medicine 2026. DOI: 10.1111/acem.70429. Observational cohort.

Procedural bleeding in cirrhosis

Rebalanced haemostasis • INR pitfalls • Procedure-specific planning

WHY “INR = BLEEDING RISK” FAILS

Cirrhosis reduces synthesis of many procoagulant factors, but also reduces natural anticoagulant proteins and alters platelets, fibrinogen and endothelial function. The resulting “rebalanced” haemostasis is fragile: bleeding and thrombosis are both possible. INR measures a limited portion of the coagulation cascade and was not validated as a procedural bleeding-risk score in cirrhosis.

START WITH THE PROCEDURE AND PATIENT

Lower-risk examples	Higher-risk examples
Paracentesis, thoracentesis	Major surgery / solid-organ biopsy
Many ultrasound-guided vascular access procedures	Selected complex endoscopic / interventional procedures

Risk depends on technical factors, vessel injury, portal hypertension, urgency, infection, acute kidney injury, degree of decompensation and whether bleeding can be controlled locally. Classification and thresholds depend on procedure-specific guidance.

WHAT NOT TO DO ROUTINELY

- Do not transfuse fresh frozen plasma solely to normalize an elevated cirrhosis-related INR before a low-risk procedure.
- Do not use a single platelet cutoff for every patient and procedure; society recommendations vary for higher-risk interventions.
- Do not ignore risks of transfusion: volume overload, portal pressure effects and transfusion reactions.

A PRACTICAL BEDSIDE PATHWAY

- Determine if there is active bleeding or an urgent high-risk procedure. Source control and resuscitation take priority when bleeding is major.
- For low-risk procedures: use good technique and ultrasound guidance when helpful; routine correction of INR and thrombocytopenia is generally not recommended.
- For higher-risk procedures: coordinate with proceduralist and hepatology/haematology; individualize platelet and fibrinogen strategies. Treat sepsis and uraemia where possible.
- For actual haemorrhage: manage physiology and haemostasis together, guided by bleeding severity, local massive haemorrhage protocol and available testing.

INTERPRETING LAB ABNORMALITIES

Lab	Meaning
INR	No universal corrective INR target for cirrhosis alone
Platelets	Threshold depends on procedure and context
Fibrinogen	Low values may matter particularly during active bleeding
TEG / ROTEM	May reduce blood use; not validated as a standalone predictor

WHAT THIS REVIEW ADDS

Davis and colleagues synthesize why pre-procedural risk assessment requires more than conventional coagulation tests. This is a clinical review, not evidence from a randomized comparison of specific thresholds.

CLINICAL PEARL

Cirrhosis with INR 2.0 is not physiologically equivalent to therapeutic warfarin anticoagulation at INR 2.0.

EVIDENCE AND READING

Davis JPE, Farias AQ, Intagliata NM. Advances in the prevention and management of procedural bleeding in patients with cirrhosis. Hepatology International 2025;19:1035–1050. DOI: 10.1007/s12072-025-10834-2. Narrative clinical review.